

William J. Baumol
Alan S. Blinder

ECONOMICS

PRINCIPLES and POLICY

Thirteenth Edition



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ECONOMICS

PRINCIPLES and POLICY

Thirteenth Edition

William J. Baumol

New York University and Princeton University

Alan S. Blinder

Princeton University



Australia • Brazil • Mexico • Singapore • United Kingdom • United States

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GETTING ACQUAINTED WITH ECONOMICS

Welcome to economics! Some of your fellow students may have warned you that “econ is boring.” Don’t believe them—or at least, don’t believe them too much. It is true that studying economics is hardly pure fun. But a first course in economics can be an eye-opening experience. There is a vast and important world out there—the economic world—and this book is designed to help you understand it.

Have you ever wondered whether jobs will be plentiful or scarce when you graduate, or why a college education becomes more and more expensive? Should the government be suspicious of big firms? Why can’t pollution be eliminated? How did the U.S. economy manage to grow so rapidly in the 1990s while Japan’s economy stagnated? If any of these questions have piqued your curiosity, read on. You may find economics is more interesting than you had thought!

It is only in later chapters that we will begin to give you the tools you need to begin carrying out your own economic analyses. However, the four chapters of Part 1 listed next will introduce you to both the subject matter of economics and some of the methods that economists use to study their subject.

- 1 What Is Economics?
- 2 The Economy: Myth and Reality
- 3 The Fundamental Economic Problem: Scarcity and Choice
- 4 Supply and Demand: An Initial Look

WHAT IS ECONOMICS?

Why does public discussion of economic policy so often show the abysmal ignorance of the participants? Why do I so often want to cry at what public figures, the press, and television commentators say about economic affairs?

ROBERT M. SOLOW, WINNER OF THE 1987 NOBEL PRIZE IN ECONOMICS

Economics is a broad-ranging discipline, both in the questions it asks and the methods it uses to seek answers. Many of the world's most pressing problems are economic in nature. The first part of this chapter is intended to give you some idea of the sorts of issues that economic analysis helps to clarify and the kinds of solutions that economic principles suggest. The second part briefly introduces the tools that economists use—tools you are likely to find useful in your career, personal life, and role as an informed citizen, long after this course is over.

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1-1 IDEAS FOR BEYOND THE FINAL EXAM

Elephants may never forget, but people do. We realize that most students inevitably forget much of what they learn in a course—perhaps with a sense of relief—soon after the final exam. Nevertheless, we hope that you will remember some of the most significant economic ideas and, even more important, the ways of thinking about economic issues that will help you evaluate the economic issues that arise in our economy.

To help you identify some of the most crucial concepts, we have selected 10 from the many in this book. Some offer key insights into the workings of the economy, and several bear on important policy issues that appear in newspapers; others point out common misunderstandings that occur among even the most thoughtful lay observers. Most of them indicate that it takes more than just good common sense to analyze economic issues effectively. As the opening quote of this chapter suggests, many learned judges, politicians, and university administrators who failed to understand basic economic principles could have made wiser decisions.

Try this one on for size. Imagine you own a widget manufacturing company that rents a warehouse. Your landlord raises your rent by \$10,000 per year. Should you raise the price of your widgets to try to recoup some of your higher costs or should you do the opposite—lower your price to try to sell more and spread the so-called overhead costs over more products? In fact, as we shall see in Chapter 8, both answers are probably wrong!



Each of the 10 “Ideas for Beyond the Final Exam,” many of which are counterintuitive, will be sketched briefly here. More important, each will be discussed in depth when it occurs in the course of the book, where it will be called to your attention by a special icon in the margin. Don’t expect to master these ideas fully now, but do notice how some of the ideas arise again and again as we deal with different topics. By the end of the course, you will have a better grasp of when common sense works and when it fails, and you will be able to recognize common fallacies that are all too often offered by public figures, the press, and television commentators.

1-1a Idea 1: How Much Does It Really Cost?

Because no one has infinite riches, people are constantly forced to make choices. If you purchase a new computer, you may have to give up that trip you had planned. If a business decides to retool its factories, it may have to postpone its plans for new executive offices. If a government expands its defense program, it may be forced to reduce its outlays on school buildings.

Economists say that the true costs of such decisions are not the number of dollars spent on the computer, the new equipment, or the military, but rather *the value of what must be given up in order to acquire the item*—the vacation trip, the new executive offices, and the new schools. These are called **opportunity costs** because they represent the opportunities the individual, firm, or government must forgo to make the desired expenditure. Economists maintain that rational decision making must be based on opportunity costs, not just dollar costs (see Chapters 3, 8, 14, and 15).

The cost of a college education provides a vivid example. How much do you think it *costs* to go to college? Most people are likely to answer by adding together their expenditures on tuition, room and board, books, and the like, and then deducting any scholarship funds they may receive. Suppose that amount comes to \$15,000.

Economists keep score differently. They first want to know how much you would be earning if you were not attending college. Suppose that salary is \$20,000 per year. This may seem irrelevant, but because you *give up* these earnings by attending college, they must be added to your tuition bill. You have that much less income because of your education. On the other side of the ledger, economists would not count *all* of the university’s bill for room and board as part of the costs of your education. They would want to know how much *more* it costs you to live at school rather than at home. Economists would count only these *extra* costs as an educational expense because you would have incurred these costs whether or not you attend college. On balance, college is probably costing you much more than you think. And, as we will see later, taking opportunity cost into account in any personal planning will help you to make more rational decisions.

1-1b Idea 2: Attempts to Repeal the Laws of Supply and Demand—The Market Strikes Back

When a commodity is in short supply, its price naturally tends to rise. Sometimes disgruntled consumers badger politicians into “solving” this problem by making the high prices illegal—by imposing a ceiling on the price. Similarly, when supplies are plentiful—say, when fine weather produces extraordinarily abundant crops—prices tend to fall. Falling prices naturally dismay producers, who often succeed in getting legislators to impose price floors.

The **opportunity cost** of a decision is the value of the next best alternative that must be given up because of that decision (e.g., working instead of going to school).

Such attempts to repeal the laws of supply and demand usually backfire and sometimes produce results virtually the opposite of those intended. Where rent controls are adopted to protect tenants, housing grows scarce because the law makes it unprofitable to build and maintain apartments. When price floors are placed under agricultural products, surpluses pile up because people buy less.

As we will see in Chapter 4 and elsewhere in this book, such consequences of interference with the price mechanism are not accidental. They follow inevitably from the way in which free markets work.

1-1c Idea 3: The Surprising Principle of Comparative Advantage

China today produces many products that Americans buy in huge quantities, including toys, textiles, and electronic equipment. American manufacturers often complain about Chinese competition and demand protection from the flood of imports that, in their view, threatens American standards of living. Is this view justified?

Economists think that it is often false. They maintain that both sides normally gain from international trade. But what if the Chinese were able to produce *everything* more cheaply than we can? Wouldn't Americans be thrown out of work and our nation be impoverished?

A remarkable result, called the law of *comparative advantage*, shows that, even in this extreme case, the two nations could still benefit by trading and that each could gain as a result! We will explain this principle first in Chapter 3 and then more fully in Chapter 35. For now, a simple parable will make the reason clear.

Suppose Sally grows up on a farm and is a whiz at plowing, but she is also a successful country singer who earns \$4,000 per performance. Should Sally turn down singing engagements to leave time to work in the fields? Of course not. Instead, she should hire Alfie, a much less efficient farmer, to do the plowing for her. Sally may be better at plowing, but she earns so much more by singing that it makes sense for her to specialize in that and leave the farming to Alfie. Although Alfie is a less skilled farmer than Sally, he is an even worse singer.

So Alfie earns his living in the job at which he at least has a *comparative* advantage (his farming is not as inferior as his singing), and both Alfie and Sally gain. The same is true of two countries. Even if one of them is more efficient at everything, both countries can gain by producing the things they do best *comparatively*.

1-1d Idea 4: Trade Is a Win–Win Situation

One of the most fundamental ideas of economics is that both parties must expect to gain something in a voluntary exchange. Otherwise, why would they both agree to trade? This principle seems self-evident, yet it is amazing how often it is ignored in practice.

For example, it was widely believed for centuries that in international trade one country's gain from an exchange must be the other country's loss (Chapter 35). Analogously, some people feel instinctively that if Ms. A profits handsomely from a deal with Mr. B, then Mr. B must have been exploited. Laws sometimes prohibit mutually beneficial exchanges between buyers and sellers—as when a loan transaction is banned because the interest rate is “too high” (Chapter 18), or when a willing worker is condemned to remain unemployed because the wage she is offered is “too low” (Chapter 19), or when the resale of tickets to sporting events (“ticket scalping”) is outlawed even though the buyer is happy to get the ticket that he could not obtain at a lower price (Chapter 4).

In every one of these cases, well-intentioned but misguided reasoning blocks the possible mutual gains that arise from voluntary exchange and thereby interferes with one of the most basic functions of an economic system (see Chapter 3).

1-1e Idea 5: The Importance of Thinking at the Margin

We will devote much of this book to explaining and extolling a type of decision-making process called *marginal analysis* (see especially Chapters 5, 7, 8, and 14), which we can best illustrate through an example.

Suppose an airline is told by its accountants that the full average cost of transporting one passenger from Los Angeles to New York is \$300. Can the airline profit by offering a reduced fare of \$200 to students who fly on a standby basis? The surprising answer is probably yes. The reason is that most of the costs of the flight must be paid whether the plane carries 20 passengers or 120 passengers.

Costs such as maintenance, landing rights, and ground crews are irrelevant to the decision of whether to carry *additional* standby passengers at reduced rates. The only costs that are relevant are the extra costs of writing and processing additional tickets, the food and beverages consumed by these passengers, the additional fuel required, and so on. These so-called *marginal costs* are probably quite small in this example. A passenger who pays the airline any amount more than it costs the airline to give her a seat that would otherwise be unused (its marginal cost of flying her) adds something to the company's profit. So it probably is more profitable to let students ride at low fares than to leave the seats empty.

In many real cases, a failure to understand marginal analysis leads decision makers to reject advantageous possibilities, like the reduced fare in our example. These people are misled by using *average* rather than *marginal* cost figures in their calculations—an error that can be very costly.

1-1f Idea 6: Externalities—A Shortcoming of the Market Cured by Market Methods

Markets are adept at producing the goods that consumers want and in just the quantities they desire. They do so by rewarding those who respond to what consumers want and who produce these commodities economically. This all works out well as long as each exchange involves only the buyer and the seller—and no one else. However, some transactions affect third parties who were not involved in the decision. Examples abound: Electric utilities that generate power for midwestern states also produce pollution that kills freshwater fish in upstate New York. A farmer sprays crops with toxic pesticides, but the poison seeps into the groundwater and affects the health of neighboring communities.

Such social costs are called *externalities* because they affect parties *external* to the economic transactions that cause them. Externalities escape the control of the market mechanism because no financial incentive motivates polluters to minimize the damage they do—as we will learn in Chapters 15 and 16. So business firms make their products as cheaply as possible, disregarding any environmental harm they may cause.

Yet Chapters 15 and 16 will point out a way for the government to use the market mechanism to control undesirable externalities. If the electric utility and the farmer are charged for the clean air and water they use, just as they are charged for any coal and fertilizer they consume, then they will have a financial incentive to reduce the amount of pollution they generate. Thus, in this case, economists believe that market methods are often the best way to cure one of the market's most important shortcomings.

1-1g Idea 7: The Trade-Off between Efficiency and Equality

Wages and income have grown more unequal in the United States since the late 1970s. Highly skilled workers have pulled away from low-skilled workers. The rich have grown richer while the poor have become (relatively) poorer, yet U.S. unemployment has been much lower than that in Europe for many years. In many European countries, inequality has not grown more extreme.

Many economists see these phenomena as two sides of the same coin. Europe and the United States have made different choices regarding how best to balance the conflicting claims of greater economic efficiency (more output and jobs) versus greater equality.

Roughly speaking, the American solution is to let markets work to promote efficiency—something they are very good at doing—with only minimal government interferences to reduce economic inequalities. (Some of these interferences are studied in Chapter 20.) However, much of continental Europe takes a different view. They find it scandalous that many Americans work for less than \$7.50 per hour, with virtually no fringe benefits and no job security. European laws mandate not only relatively high minimum wages but also substantial fringe benefits and employment protections; of course, European taxes must be much higher to pay for these programs.

As economists see it, each system's virtue is also its vice. There is an agonizing *trade-off* between the *size* of a nation's output and the degree of *equality* with which that output is distributed. European-style policies designed to divide the proverbial economic pie more equally inadvertently can cause the size of the pie to shrink. American-style arrangements that promote maximal efficiency and output may permit or even breed huge inequalities and poverty. Which system is better? There is no clear answer, but we will examine the issue in detail in Chapter 20.

1-1h Idea 8: Government Policies Can Limit Economic Fluctuations—But Don't Always Succeed

One of the most persistent and troubling problems of market economies has been their tendency to go through cycles of boom and bust. The booms, as we shall see, often bring inflation, and the busts always raise unemployment. Years ago, economists, businesspeople, and politicians viewed these fluctuations as inevitable: There was nothing the government could or should do about them.

That view is now considered obsolete. As we will learn in Part 7, and especially Part 8, modern governments have an arsenal of weapons that they can and do deploy to try to mitigate fluctuations in their national economies—to limit both inflation and unemployment. Some of these weapons constitute what is called *fiscal policy*: control over taxes and government spending. Others come from *monetary policy*: control over money and interest rates. Both were used on a grand scale to fight the Great Recession of 2007–2009.

But *trying* to tame the business cycle is not the same as *succeeding*. Economic fluctuations remain with us, and one reason is that the government's fiscal and monetary policies sometimes fail—for both political and economic reasons. As we will see in Part 8, policy makers do not always make the right decisions. And even when they do, the economy does not always react as policymakers hope. Furthermore, for reasons we will explain later, the “right” decision is not always clear. To this day, many of the fiscal and monetary policies of 2008 and 2009 remain highly controversial.

1-1i Idea 9: The Short-Run Trade-Off between Inflation and Unemployment

The U.S. economy was lucky in the second half of the 1990s. A set of fortuitous events—falling energy prices, tumbling computer prices, a rising dollar, and so on—pushed inflation down even as unemployment fell to its lowest level in almost 30 years. During the 1970s and early 1980s, the United States was not so fortunate. Skyrocketing prices for food and energy sent both inflation and unemployment up to extraordinary heights. In both episodes, then, inflation and unemployment moved in the same direction.

But economists maintain that neither of these two episodes was “normal.” When we are experiencing neither unusually good luck (as in the 1990s) nor exceptionally bad luck (as in the 1970s), there is a *trade-off between inflation and unemployment*—meaning that low

unemployment normally makes inflation rise and high unemployment normally makes inflation fall. For example, the high unemployment of 2008–2010 pushed the inflation rate down so low that people began worrying about *negative* inflation rates, or *deflation*. We will study the mechanisms underlying this trade-off in Parts 7 and 8, especially in Chapter 34. It poses one of the fundamental dilemmas of national economic policy.

1-1j **Idea 10: Productivity Growth Is (Almost) Everything in the Long Run**

Today in Geneva, Switzerland, workers in a watch factory turn out more than 100 times as many mechanical watches per year as their ancestors did three centuries earlier. The productivity of labor (output per hour of work) in cotton production has probably gone up more than 1,000-fold in 200 years. It is estimated that rising labor productivity has increased the standard of living of a typical American worker approximately sevenfold in the past century (see Chapter 24).

Other economic issues such as unemployment, monopoly, and inequality are important to us all and receive much attention in economics. But in the long run, nothing has as great an effect on our material well-being and the amounts society can afford to spend on hospitals, schools, and social amenities as the rate of growth of productivity—the amount that an average worker can produce in an hour. Chapter 24 points out that what appears to be a small increase in productivity growth can have a huge effect on a country's standard of living over a long period of time because productivity compounds like the interest on savings in a bank. Similarly, a slowdown in productivity growth that persists for a substantial number of years can have a devastating effect on living standards.

1-1k **Epilogue**

These ideas are some of the more fundamental concepts you will find in this book—ideas that we hope you will retain beyond the final exam. There is no need to master them right now, for you will hear much more about each as you progress through the book. By the end of the course, you may be amazed to see how natural, or even obvious, they will seem.

1-2 **INSIDE THE ECONOMIST'S TOOL KIT**

We turn now from the kinds of issues economists deal with to some of the tools they use to grapple with them.

1-2a **Economics as a Discipline**

Although economics is clearly the most rigorous of the social sciences, it nevertheless looks decidedly more “social” than “scientific” when compared with, say, physics. An economist must be a jack of several trades, borrowing modes of analysis from numerous fields. Mathematical reasoning is often used in economics, but so is historical study. And neither looks quite the same as when practiced by a mathematician or a historian. Statistics play a major role in modern economic inquiry, although economists had to modify standard statistical procedures to fit their kinds of data.

1-2b **The Need for Abstraction**

Some students find economics unduly abstract and “unrealistic.” The stylized world envisioned by economic theory seems only a distant cousin to the world they know. There is an old joke about three people—a chemist, a physicist, and an economist—stranded on a desert island with an ample supply of canned food but no tools to open the cans. The chemist thinks that lighting a fire under the cans would burst the cans. The physicist advocates building a catapult with which to smash the cans against some boulders. The economist's suggestion? “Let's assume we have a can opener.”

Economic theory *does* make some unrealistic assumptions—you will encounter some of them in this book—but some abstraction from reality is necessary because of the incredible complexity of the economic world, not because economists like to sound absurd.

Compare the chemist's simple task of explaining the interactions of compounds in a chemical reaction with the economist's complex task of explaining the interactions of people in an economy. Are molecules motivated by greed or altruism, by envy or ambition? Do they ever imitate other molecules? Do forecasts about them influence their behavior? People, of course, do all these things and many, many more. It is therefore vastly more difficult to predict human behavior than to predict chemical reactions. If economists tried to keep track of every feature of human behavior, they would never get anywhere. Thus:

Abstraction from unimportant details is necessary to understand the functioning of anything as complex as the economy.

An analogy will make it clear why economists **abstract** from details. Suppose you have just arrived for the first time in Los Angeles. You are now at the Los Angeles Civic Center—the point marked *A* in Maps 1 and 2, which are alternative maps of part of Los Angeles. You want to drive to the Los Angeles County Museum of Art, point *B* on each map. Which map would be more useful?

Map 1 has complete details of the Los Angeles road system, but this makes it hard to read and hard to use as a way to find the art museum. For this purpose, Map 1 is far too detailed, although for other purposes (e.g., locating a small street in Hollywood) it may be far better than Map 2.

In contrast, Map 2 omits many minor roads—you might say they are *assumed away*—so that the freeways and major arteries stand out more clearly. As a result of this simplification, several routes from the Civic Center to the Los Angeles County Museum of Art

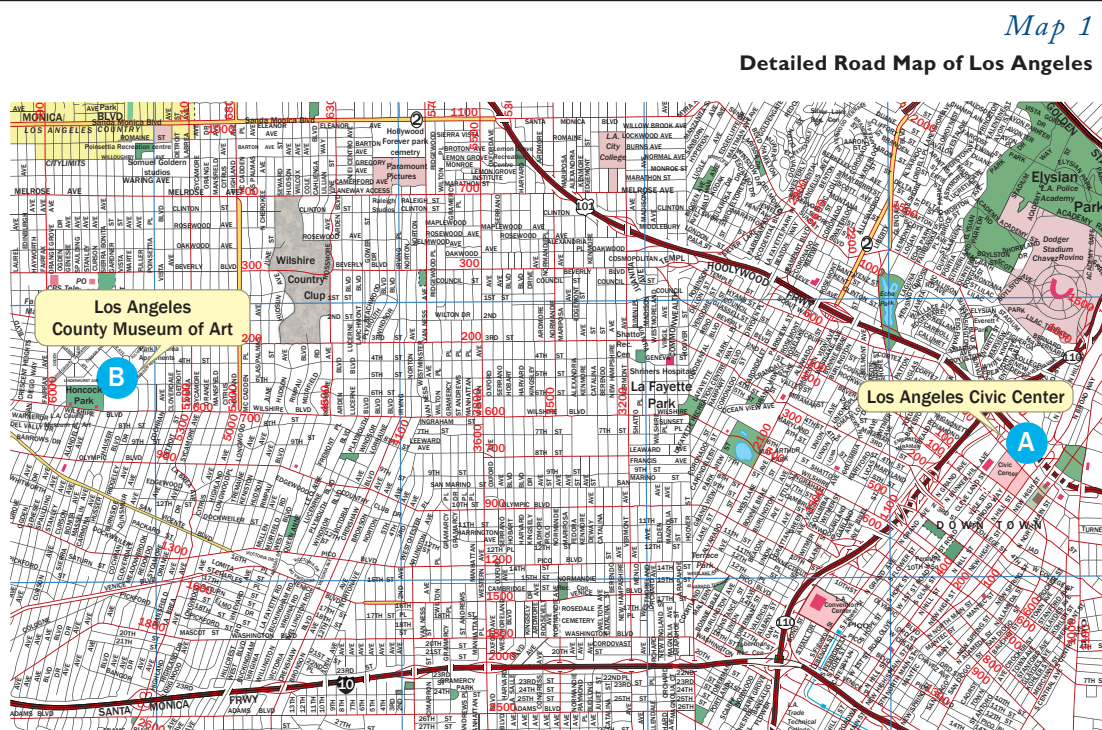


From The Wall Street Journal—Permission, Cartoon Features Syndicate.

“Yes, John, we’d all like to make economics less dismal ...”

NOTE: The nineteenth-century British writer Thomas Carlyle described economics as the “dismal science,” a label that stuck.

Abstraction means ignoring many details so as to focus on the most important elements of a problem.



NOTE: Point A marks the Los Angeles Civic Center, and point B marks the Los Angeles County Museum of Art.

Map 2

Major Los Angeles Arteries and Freeways



emerge. For example, we can take the Hollywood Freeway west to Alvarado Boulevard, go south to Wilshire Boulevard, and then head west again. Although we *might* find a shorter route by poring over the details in Map 1, most strangers to the city would be better off with Map 2. Similarly, economists try to *abstract* from a lot of confusing details while retaining the essentials.

Map 3, however, illustrates that simplification can go too far. It shows little more than the major interstate routes that pass through the greater Los Angeles area and therefore will not help a visitor find the art museum. Of course, this map was never intended to be used as a detailed tourist guide, which brings us to an important point:

There is no such thing as one “right” degree of abstraction and simplification for all analytic purposes. The proper degree of abstraction depends on the objective of the analysis. A model that is a gross oversimplification for one purpose may be needlessly complicated for another.

Economists are constantly seeking analogies to Map 2 rather than Map 3, walking the thin line between useful generalizations about complex issues and gross distortions of the pertinent facts. For example, suppose you want to learn why some people are fabulously rich whereas others are abjectly poor. People differ in many ways, too many to enumerate, much less to study. The economist must ignore most of these details to focus on the important ones. The color of a person’s hair or eyes is probably not important for the problem but, unfortunately, the color of his or her skin probably is because racial discrimination can depress a person’s income. Height and weight may not matter, but education probably does. Proceeding in this way, we can pare Map 1 down to the manageable dimensions of Map 2. But there is a danger of going too far, stripping away some of the crucial factors, so that we wind up with Map 3.

Map 3

Greater Los Angeles Freeways



1-2c The Role of Economic Theory

Some students find economics “too theoretical.” To see why we can’t avoid it, let’s consider what we mean by a **theory**.

To an economist or natural scientist, the word *theory* means something different from what it means in common speech. In science, a theory is *not* an untested assertion of alleged fact. The statement that aspirin provides protection against heart attacks is not a theory; it is a *hypothesis*, that is, a reasoned guess, which will prove to be true or false once the right sorts of experiments have been completed. But a theory is different. It is a deliberate simplification (abstraction) of reality that attempts to explain how some relationships work. It is an *explanation* of the mechanism behind observed phenomena. Thus, gravity forms the basis of theories that describe and explain the paths of the planets. Similarly, Keynesian theory (discussed in Parts 7 and 8) seeks to describe and explain how government policies affect unemployment and prices in the national economy.

People who have never studied economics often draw a false distinction between *theory* and *practical policy*. Politicians and businesspeople, in particular, often reject abstract economic theory as something that is best ignored by “practical” people. The irony of these statements is that

It is precisely the concern for policy that makes economic theory so necessary and important.

To analyze policy options, economists are forced to deal with *possibilities that have not actually occurred*. For example, to learn how to shorten periods of high unemployment, they must investigate whether a proposed new policy that has never been tried can help. Or to determine

A **theory** is a deliberate simplification of relationships used to explain how those relationships work.

Two variables are said to be **correlated** if they tend to go up or down together. Correlation need not imply causation.

which environmental programs will be most effective, they must understand how and why a market economy produces pollution and what might happen if the government taxed industrial waste discharges and automobile emissions. Such questions require some *theorizing*, not just examination of the facts, because we need to consider possibilities that have never occurred.

The facts, moreover, can sometimes be highly misleading. Data often indicate that two variables move up and down together. But this statistical **correlation** does not prove that either variable *causes* the other. For example, when it rains, people drive slower and there are also more traffic accidents, but no one thinks slower driving causes more accidents. Rather, we understand that both phenomena are caused by a common underlying factor—more rain. How do we know this? Not just by looking at the correlation between data on accidents and driving speeds. Data alone tell us little about cause and effect. We must use some simple *theory* as part of our analysis. In this case, the theory might explain that drivers are more apt to have accidents on wet roads.

Similarly, we must use theoretical analysis, and not just data alone, to understand *how*, if at all, different government policies will lead to lower unemployment or *how* a tax on emissions will reduce pollution.

Statistical correlation need not imply causation. Some theory is usually needed to interpret data.

1-2d What Is an Economic Model?

An **economic model** is a simplified, small-scale version of an aspect of the economy. Economic models are often expressed in equations, by graphs, or in words.

An **economic model** is a representation of a theory or a part of a theory, often used to gain insight into cause and effect. The notion of a “model” is familiar enough to children, and economists—like other researchers—use the term the same way children do.

A child’s model airplane looks and operates much like the real thing, but it is smaller and simpler, so it is easier to manipulate and understand. Engineers for Boeing also build models of planes. Although their models are far larger and much more elaborate than a child’s toy, they use them for the same purposes: to observe the workings of these aircraft “up close” and to experiment to see how the models behave under different circumstances. (“What happens if I do this?”) From these experiments, they make educated guesses as to how the real-life version will perform.

Economists use models for similar purposes. The late A. W. Phillips, famous engineer-turned-economist who discovered the “Phillips curve” (discussed in Chapter 34), was talented enough to construct a working model of the determination of national income in a simple economy by using colored water flowing through pipes. For years this contraption has graced the basement of the London School of Economics. Although we will explain the models with words and diagrams, Phillips’s engineering background enabled him to depict the theory with tubes, valves, and pumps.

Because many of the models used in this book are depicted in diagrams, for those of you who need review, we explain the construction and use of various types of graphs in the appendix to this chapter. Don’t be put off by seemingly abstract models. Think of them as useful road maps and remember how hard it would be to find your way around Los Angeles without one.



Science & Society Picture Library/Getty Images

A. W. Phillips built this model in the early 1950s to illustrate Keynesian theory.

1-2e Reasons for Disagreements: Imperfect Information and Value Judgments

“If all the earth’s economists were laid end to end, they could not reach an agreement,” the saying goes. Politicians and reporters are fond of pointing out that economists can be found on both sides of many public policy issues. If economics is a science, why do economists so often disagree? After all, astronomers do not debate whether the earth revolves around the sun or vice versa.

This question reflects a misunderstanding of the nature of science. Disputes are normal at the frontier of any science. For example, astronomers once argued vociferously over whether the earth revolves around the sun. Nowadays, they argue about gamma-ray bursts, dark matter, and other esoterica. These arguments go mostly unnoticed by the public because few of us understand what they are talking about. But economics is a *social* science, so its disputes are aired in public and all sorts of people feel competent to join economic debates.

Furthermore, economists actually agree on much more than is commonly supposed. Virtually all economists, regardless of their politics, agree that taxing polluters is one of the best ways to protect the environment (see Chapters 15 and 16), that rent controls can ruin a city (Chapter 4), and that free trade among nations is usually preferable to the erection of barriers through tariffs and quotas (see Chapter 35). The list could go on and on. It is probably true that the issues about which economists agree *far* exceed the subjects on which they disagree.

Finally, many disputes among economists are not scientific disputes at all. Sometimes the pertinent facts are simply unknown. For example, you will learn in Chapter 16 that the appropriate financial penalty to levy on a polluter depends on quantitative estimates of the harm done by the pollutant; however, good estimates of this damage may not be available. Similarly, although there is wide scientific agreement that the earth is slowly warming, there are disagreements over the costs of global warming. Such disputes make it difficult to agree on a concrete policy proposal.

Another important source of disagreements is that economists, like other people, come in all political stripes: conservative, middle-of-the-road, liberal, radical. Each may have different values, and so each may hold a different view of the “right” solution to a public policy problem—even if they agree on the underlying analysis. Here are two examples:

1. We suggested early in this chapter that policies that lower inflation are likely to raise unemployment. Many economists believe they can measure the amount of unemployment that must be endured to reduce inflation by a given amount. However, they disagree about whether it is worth having, say, 3 million more people out of work for a year to cut the inflation rate by 1 percent.
2. In designing an income tax, society must decide how much of the burden to put on upper-income taxpayers. Some people believe the rich should pay a disproportionate share of the taxes. Others disagree, believing it is fairer to levy the same income-tax rate on everyone.

Economists cannot answer questions like these any more than nuclear physicists could have determined whether dropping the atomic bomb on Hiroshima was a good idea. The decisions rest on moral judgments that can be made only by the citizenry through its elected officials.

Although economic science can contribute theoretical and factual knowledge on a particular issue, the final decision on policy questions often rests either on information that is not currently available or on social values and ethical opinions about which people differ, or on both.

Summary

1. To help you get the most out of your first course in economics, we have devised a list of 10 important ideas that you will want to retain beyond the final exam. Briefly, they are the following:
 - a. **Opportunity cost** is the correct measure of cost.
 - b. Attempts to fight market forces often backfire.
 - c. Nations can gain from trade by exploiting their *comparative advantages*.
 - d. Both parties can gain in a voluntary exchange.
 - e. Good decisions typically require *marginal analysis*, which weighs added costs against added benefits.
 - f. Externalities may cause the market mechanism to malfunction, but this defect can often be repaired by market methods.
 - g. Governments have tools that can mitigate cycles of boom and bust, but these tools are imperfect.
 - h. There is a trade-off between efficiency and equality. Many policies that promote one damage the other.

- i. In the short run, policy makers face a trade-off between inflation and unemployment. Policies that reduce one normally increase the other.
- j. In the long run, *productivity* is almost the only thing that matters for a society's material well-being.
2. Common sense is not always a reliable guide in explaining economic issues or in making economic decisions.
3. Because of the great complexity of human behavior, economists are forced to **abstract** from many details, to make generalizations that they know are not quite true, and to organize what knowledge they have in terms of some theoretical structure called a "model."
4. **Correlation** need not imply causation.
5. Economists use simplified models to understand the real world and predict its behavior, much as a child uses a model railroad to learn how trains work.
6. Although these **economic models**, if skillfully constructed, can illuminate important economic problems, they rarely can answer the questions that confront policy makers. Value judgments involving such matters as ethics are needed for this purpose, and the economist is no better equipped than anyone else to make them.

Key Terms

abstraction 9

correlation 12

economic model 12

opportunity cost 4

theory 11

Discussion Questions

1. Think about a way you would construct a model of how your college is governed. Which officers and administrators would you include and exclude from your model if the objective were one of the following:
 - a. To explain how decisions on financial aid are made
 - b. To explain the quality of the faculty
 Relate this to the map example in the chapter.
2. Relate the process of abstraction to the way you take notes in a lecture. Why do you not try to transcribe every word uttered by the lecturer? Why don't you write down just the title of the lecture and stop there? How do you decide, roughly speaking, on the correct amount of detail?
3. Explain why a government policy maker cannot afford to ignore economic theory.

Appendix Using Graphs: A Review¹

As noted in the chapter, economists often explain and analyze models with the help of graphs. Indeed, this book is full of them. But that is not the only reason for studying how graphs work. Most college students will deal with graphs in the future, perhaps frequently. You will see them in newspapers. If you become a doctor, you will use graphs to keep track of your patients' progress. If you join a business firm, you will use them to check profit or performance at a glance. This appendix introduces some of the techniques of graphic analysis—tools you will use throughout the book and, more important, very likely throughout your working career.

Graphs Used in Economic Analysis

Economic graphs are invaluable because they can display a large quantity of data quickly and because they facilitate data interpretation and analysis. They enable

the eye to take in important statistical relationships at a glance that would be far less apparent from written descriptions or long lists of numbers.

Two-Variable Diagrams

Much of the economic analysis found in this and other books requires that we keep track of two **variables** simultaneously.

A variable is something measured by a number; it is used to analyze what happens to other things when the size of that number changes (varies).

For example, in studying how markets operate, we will want to keep one eye on the *price* of a commodity and the other on the *quantity* of that commodity that is bought and sold.

For this reason, economists frequently find it useful to display real or imaginary figures in a two-variable diagram, which simultaneously represents the behavior of two economic variables. The numerical value

¹Students who have some acquaintance with geometry and feel quite comfortable with graphs can safely skip this appendix.

of one variable is measured along the horizontal line at the bottom of the graph (called the *horizontal axis*), starting from the **origin** (the point labeled “0”), and the numerical value of the other variable is measured up the vertical line on the left side of the graph (called the *vertical axis*), also starting from the origin.

The “0” point in the lower-left corner of a graph where the axes meet is called the origin. Both variables are equal to zero at the origin.

Figures 1(a) and 1(b) are typical graphs of economic analysis. They depict an imaginary *demand curve*, represented by the blue dots in Figure 1(a) and the heavy blue line in Figure 1(b). The graphs show the price of natural gas on their vertical axes and the quantity of gas people want to buy at each price on the horizontal axes. The dots in Figure 1(a) are connected by the continuous blue curve labeled *DD* in Figure 1(b).

Economic diagrams are generally read just as one would read latitudes and longitudes on a map. On the demand curve in Figure 1, the point marked *a* represents a hypothetical combination of price and quantity of natural gas demanded by customers in St. Louis. By drawing a horizontal line leftward from that point to the vertical axis, we learn that at this point the average price for gas in St. Louis is \$3 per thousand cubic feet. By dropping a line straight down to the horizontal axis, we find that consumers want 80 billion cubic feet per year at this price, just as the statistics in Table 1 show. The other points on the graph give similar information. For example, point *b* indicates that if natural gas in St. Louis were to cost only \$2 per thousand cubic feet, quantity demanded would be higher—it would reach 120 billion cubic feet per year.

Notice that information about price and quantity is *all* we can learn from the diagram. The demand curve will not tell us what kinds of people live in

Table 1

Quantities of Natural Gas Demanded at Various Prices

Price (per thousand cubic feet)	\$2	\$3	\$4	\$5	\$6
Quantity demanded (billions of cubic feet per year)	120	80	56	38	20

St. Louis, the size of their homes, or the condition of their furnaces. It tells us about the quantity demanded at each possible price—no more, no less.

A diagram abstracts from many details, some of which may be quite interesting, so as to focus on the two variables of primary interest—in this case, the price of natural gas and the amount of gas that is demanded at each price. All of the diagrams used in this book share this basic feature. They cannot tell the reader the “whole story,” any more than a map’s latitude and longitude figures for a particular city can make someone an authority on that city.

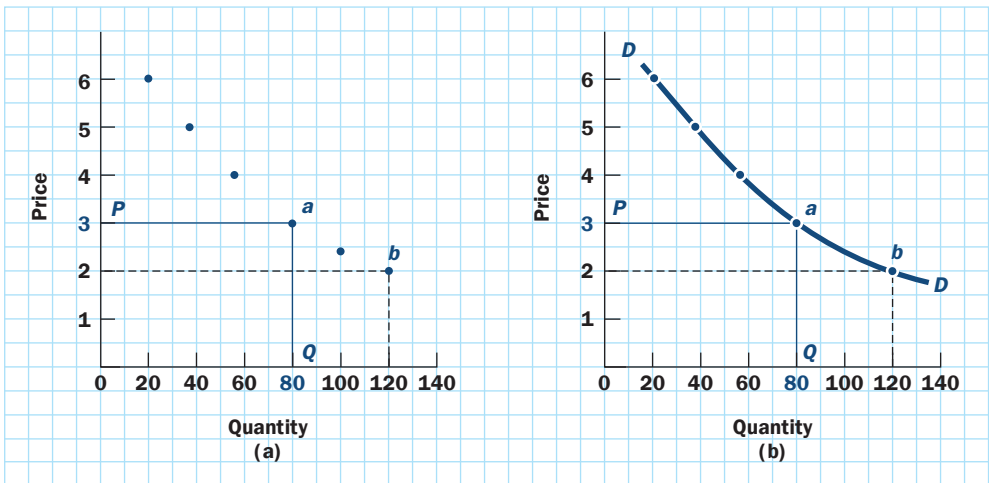
The Definition and Measurement of Slope

One of the most important features of economic diagrams is the rate at which the line or curve being sketched runs uphill or downhill as we move to the right. The demand curve in Figure 1 clearly slopes downhill (the price falls) as we follow it to the right (i.e., as consumers demand more gas). In such instances, we say that *the curve has a negative slope, or is negatively sloped, because one variable falls as the other one rises.*

The slope of a straight line is the ratio of the vertical change to the corresponding horizontal change as we move to the right along the line between two points on that line, or, as it is often said, the ratio of the “rise” over the “run.”

Figure 1

A Hypothetical Demand Curve for Natural Gas in St. Louis



NOTE: Price is in dollars per thousand cubic feet; quantity is in billions of cubic feet per year.

The four panels of Figure 2 show all possible types of slope for a straight-line relationship between two unnamed variables called Y (measured along the vertical axis) and X (measured along the horizontal axis). Figure 2(a) shows a *negative slope*, much like our demand curve in the previous graph. Figure 2(b) shows a *positive slope*, because variable Y rises (we go uphill) as variable X rises (as we move to the right). Figure 2(c) shows a *zero slope*, where the value of Y is the same irrespective of the value of X . Figure 2(d) shows an *infinite slope*, meaning that the value of X is the same irrespective of the value of Y .

Slope is a numerical concept, not just a qualitative one. The two panels of Figure 3 show two positively sloped straight lines with different slopes. The line in Figure 3(b) is clearly steeper. But by how much? The labels should help you compute the answer. In Figure 3(a) a horizontal movement, AB , of 10 units ($13 - 3$) corresponds to a vertical movement, BC , of 1 unit ($9 - 8$). So the slope is $BC/AB = 1/10$. In Figure 3(b), the same horizontal movement of 10 units corresponds to a vertical movement of 3 units ($11 - 8$). So the slope is $3/10$, which is larger—the rise divided by the run is greater in Figure 3(b).

By definition, the slope of any particular straight line remains the same, no matter where on that line

we choose to measure it. That is why we can pick any horizontal distance, AB , and the corresponding slope triangle, ABC , to measure slope. But this is not true for curved lines.

Curved lines also have slopes, but the numerical value of the slope differs at every point along the curve as we move from left to right.

The four panels of Figure 4 provide some examples of **slopes of curved lines**. The curve in Figure 4(a) has a negative slope everywhere, and the curve in Figure 4(b) has a positive slope everywhere. But these are not the only possibilities. In Figure 4(c) we encounter a curve that has a positive slope at first but a negative slope later on. Figure 4(d) shows the opposite case: a negative slope followed by a positive slope.

We can measure the slope of a smooth curved line numerically *at any particular point* by drawing a *straight line that touches, but does not cut, the curve at the point in question*. Such a line is called a **tangent to the curve**.

The slope of a curved line at a particular point is defined as the slope of the straight line that is tangent to the curve at that point.

Figure 2

Different Types of Slope of a Straight-Line Graph

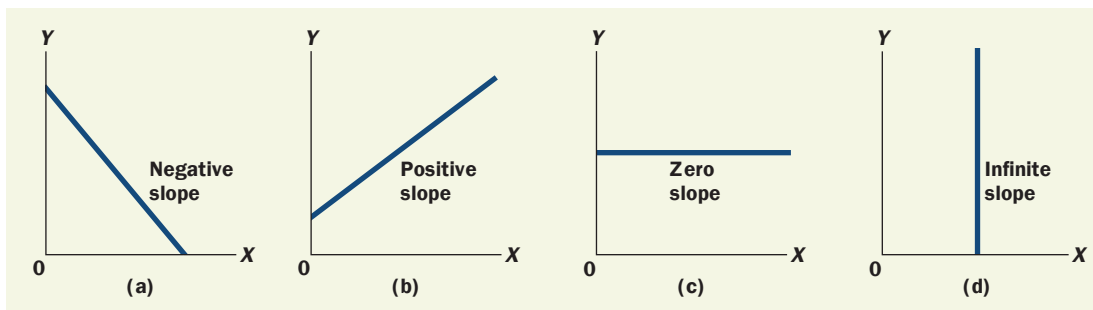


Figure 3

How to Measure Slope

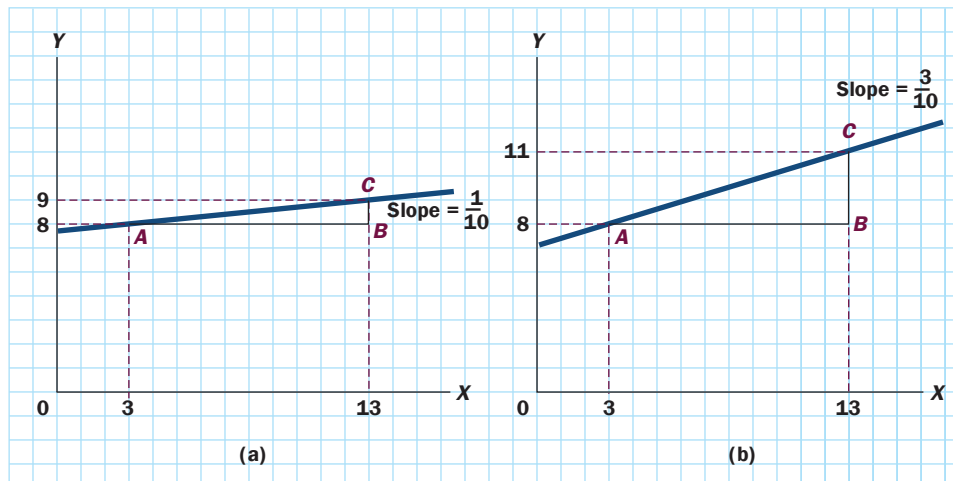


Figure 4

Behavior of Slopes in Curved Graphs

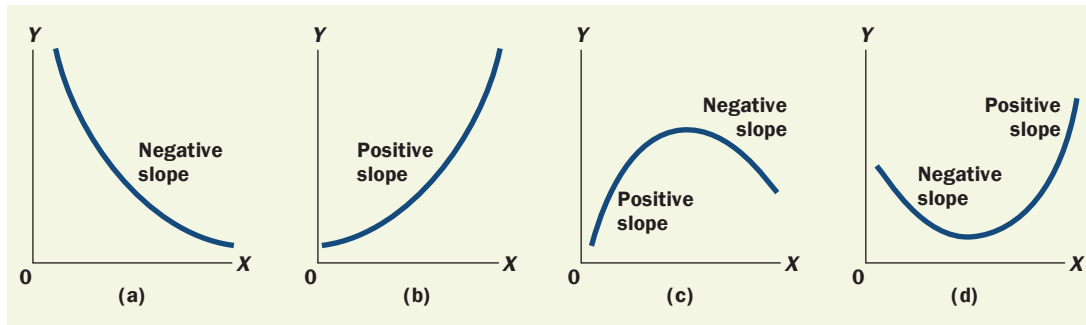


Figure 5

How to Measure Slope at a Point on a Curved Graph

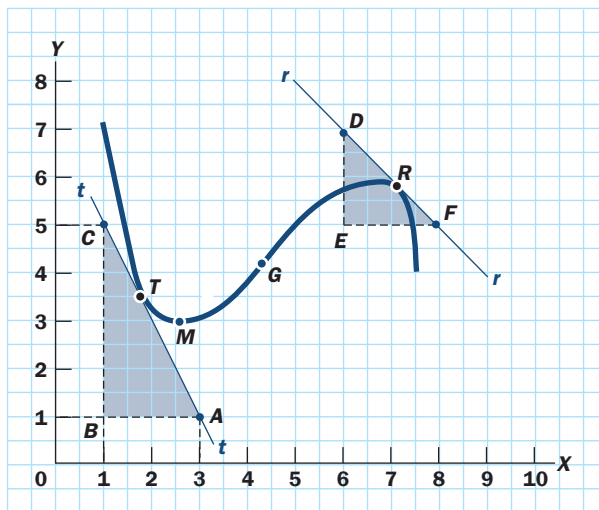


Figure 5 shows tangents to the blue curve at two points. Line tt is tangent at point T , and line rr is tangent at point R . We can measure the slope of the curve at these two points by applying the definition. The calculation for point T , then, is the following:

$$\begin{aligned}\text{Slope at point } T &= \text{Slope of line } tt \\ &= \frac{\text{Distance } BC}{\text{Distance } BA} \\ &= \frac{(5 - 1)}{(3 - 1)} = \frac{4}{2} = 2\end{aligned}$$

A similar calculation yields the slope of the curve at point R , which, as we can see from Figure 5, must be smaller numerically. That is, the tangent line rr is less steep than line tt :

$$\begin{aligned}\text{Slope at point } R &= \text{Slope of line } rr \\ &= \frac{(7 - 5)}{(8 - 6)} = \frac{2}{2} = 1\end{aligned}$$

Exercise Show that the slope of the curve at point G is about 1.

What would happen if we tried to apply this graphical technique to the high point in Figure 4(c) or to the low point in Figure 4(d)? Take a ruler and try it. The tangents that you construct should be horizontal, meaning that they should have a slope exactly equal to zero. It is always true that where the slope of a *smooth* curve changes from positive to negative, or vice versa, there will be at least one point whose slope is zero.

Curves shaped like smooth hills, as in Figure 4(c), have a zero slope at their *highest* point. Curves shaped like valleys, as in Figure 4(d), have a zero slope at their *lowest* point.

Rays through the Origin and 45° Lines

The point at which a straight line cuts the vertical (Y) axis is called the **Y -intercept**.

The Y -intercept of a line or a curve is the point at which it touches the vertical axis (the Y -axis). The X -intercept is defined similarly.

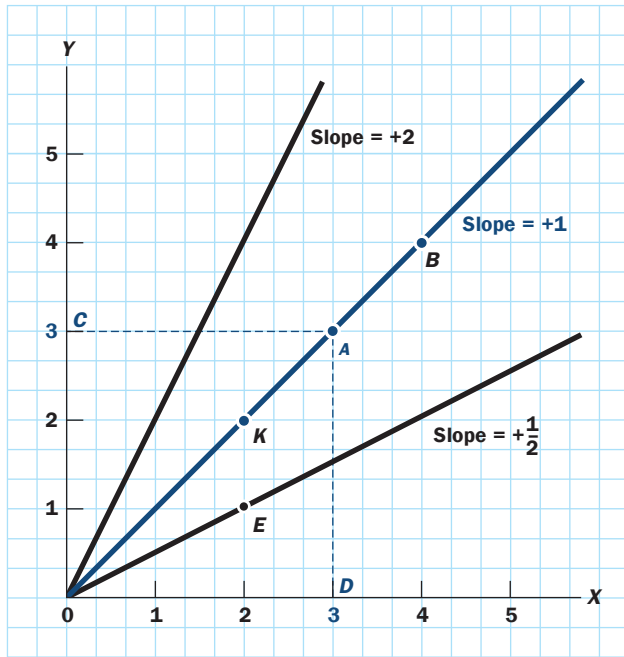
For example, the Y -intercept of the line in Figure 3(a) is a bit less than 8.

Lines whose Y -intercept is zero have so many special uses in economics and other disciplines that they have been given a special name: a ray through the origin, or a ray.

Figure 6 shows three rays through the origin, and the slope of each is indicated in the diagram. The ray in the center (whose slope is 1) is particularly useful in many economic applications because it marks points where X and Y are equal (as long as X and Y are measured in the same units). For example, at point A we have $X = 3$ and $Y = 3$; at point B , $X = 4$ and $Y = 4$. A similar relation holds at any other point on that ray.

How do we know that this is always true for a ray whose slope is 1? If we start from the origin (where

Figure 6
Rays through the Origin



both X and Y are zero) and the slope of the ray is 1, we know from the definition of slope that

$$\text{Slope} = \frac{\text{Vertical change}}{\text{Horizontal change}} = 1$$

This implies that the vertical change and the horizontal change are always equal, so the two variables must always remain equal. Any point along that ray (e.g., point A) is exactly equal in distance from the horizontal and vertical axes (length $DA = \text{length } CA$)—the number on the X -axis (the abscissa) will be the same as the number on the Y -axis (the ordinate).

Rays through the origin with a slope of 1 are called 45° lines because they form an angle of 45° with the horizontal axis. A 45° line marks off points where the variables measured on each axis have equal values.²

If a point representing some data is above the 45° line, we know that the value of Y exceeds the value of X . Similarly, whenever we find a point below the 45° line, we know that X is larger than Y .

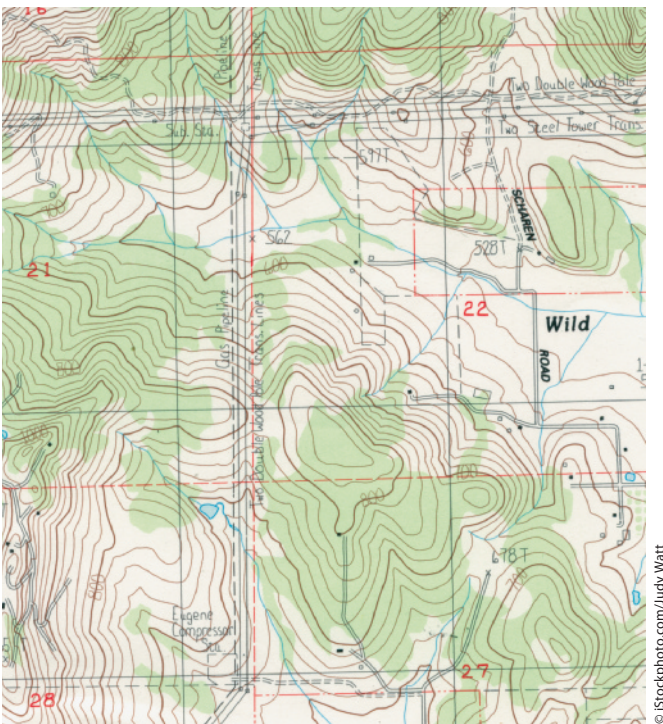
Squeezing Three Dimensions into Two: Contour Maps

Sometimes problems involve more than two variables, so two dimensions just are not enough to depict them on a graph. This is unfortunate, because the surface of a sheet of paper is only two-dimensional. When we study a business firm's decision-making process, for example, we may want to keep track simultaneously of three variables: how much labor it employs, how much raw material it imports from foreign countries, and how much output it creates.

Luckily, economists can use a well-known device for collapsing three dimensions into two—a *contour map*. Figure 7 is a contour map of the area just southeast of Eugene, Oregon, including the Eugene Compressor Station. On some of the irregularly shaped “rings” on this map, we find numbers (like 800) indicating the height (in meters) above sea level at that particular spot on the hill. Thus, unlike other maps, which give only latitudes and longitudes, this contour map (also called a topographical map) exhibits *three* pieces of information about each point: latitude, longitude, and altitude.

Figure 8 looks more like the contour maps encountered in economics. It shows how a third variable, called Z (think of it as a firm's output, for example), varies as we change either variable X (think of it as a firm's employment of labor) or variable Y (think of it as the use of imported raw material). Just like the map of the area near Eugene, Oregon, any point on

Figure 7
A Geographic Contour Map

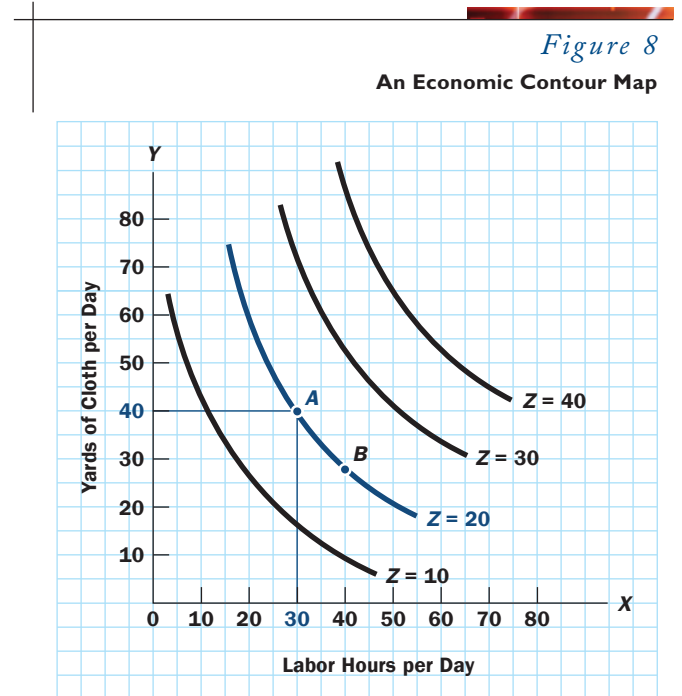


² The definition assumes that both variables are measured in the same units.

the diagram conveys three pieces of data. At point *A*, we can read off the values of *X* and *Y* in the conventional way (*X* is 30 and *Y* is 40), and we can also note the value of *Z* by finding out on which contour line point *A* falls. (It is on the $Z = 20$ contour.) So point *A* is able to tell us that 30 hours of labor and 40 yards of cloth produce 20 units of output per day. The contour line that indicates 20 units of output shows the various combinations of labor and cloth a manufacturer can use to produce 20 units of output. Economists call such maps **production indifference maps**.

A production indifference map is a graph whose axes show the quantities of two inputs that are used to produce some output. A curve in the graph corresponds to some given quantity of that output, and the different points on that curve show the different quantities of the two inputs that are just enough to produce the given output.

Although most of the analyses presented in this book rely on the simpler two-variable diagrams, contour maps will find their applications, especially in the appendixes to Chapters 5 and 7.



Summary

1. Because graphs are used so often to portray economic models, it is important for students to acquire some understanding of their construction and use. Fortunately, the graphics used in economics are usually not very complex.
2. Most economic models are depicted in two-variable diagrams. We read data from these diagrams just as we read the latitude and longitude on a map: each point represents the values of two variables at the same time.
3. In some instances, three variables must be shown at once. In these cases, economists use contour maps, which, as the name suggests, show "latitude," "longitude," and "altitude" all at the same time.
4. Often, the most important property of a line or curve drawn on a diagram will be its slope, which is defined as the ratio of the "rise" over the "run," or the vertical change divided by the horizontal change when one moves along the curve. Curves that go uphill as we move to the right have positive slopes; curves that go downhill have negative slopes.
5. By definition, a straight line has the same slope wherever we choose to measure it. The slope of a curved line changes, but the slope at any point on the curve can be calculated by measuring the slope of a straight line tangent to the curve at that point.

Key Terms

45° line 18
origin (of a graph) 15
production indifference map 19

ray through the origin, or ray 17
slope of a straight (or curved) line 15–16

tangent to a curve 16
variable 14
Y-intercept 17

Test Yourself

1. Portray the following hypothetical data on a two-variable diagram:

Academic Year	Total Enrollment	Enrollment in Economics Courses
2008–2009	3,000	300
2009–2010	3,100	325
2010–2011	3,200	350
2011–2012	3,300	375
2012–2013	3,400	400

Measure the slope of the resulting line, and explain what this number means.

2. From Figure 5, calculate the slope of the curve at point *M*.
3. Colin believes that the number of job offers he will get depends on the number of courses in which his

grade is B+ or better. He concludes from observation that the following figures are typical:

Number of grades of B+ or better	0	1	2	3	4
Number of job offers	1	3	4	5	6

Put these numbers into a graph like Figure 1(a). Measure and interpret the slopes between adjacent dots.

4. In Figure 6, determine the values of *X* and *Y* at point *K* and at point *E*. What do you conclude about the slopes of the lines on which *K* and *E* are located?
5. In Figure 8, interpret the economic meaning of points *A* and *B*. What do the two points have in common? What is the difference in their economic interpretation?

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